CS-E4740 - Federated Learning '26

Project Kickoff

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Calendar **Glossary** **Book** **GitHub**

Table of Contents

Mid-Course Feedback

Getting Started with FMI Data

The 3-Step Workflow

Validating Your Results

Deliverables and Grading

10 ECTS Extension

Next Steps

Table of Contents

Mid-Course Feedback

Getting Started with FMI Data

The 3-Step Workflow

Validating Your Results

Deliverables and Grading

10 ECTS Extension

Next Steps

Feedback Highlights

Respondents: 32 / 106 (30%)

Time spent so far	59%	around 50 h
Workload	75%	as expected
(GTVMin formulation)	87%	confident / mostly there
(fixed-point equation)	81%	confident / mostly there
(FL algorithm design)	83%	confident / mostly there

Least clear: Step 3 (46%), Step 2 (34%)

Kickoff priorities:

Choosing / implementing an FL algorithm	34%
Getting started with FMI data	33%
Grading and peer review	16%
High-quality / publishable report	16%

Table of Contents

Mid-Course Feedback

Getting Started with FMI Data

The 3-Step Workflow

Validating Your Results

Deliverables and Grading

10 ECTS Extension

Next Steps

FMI Data

Raw data source (fixed for all projects): FMI Open Data

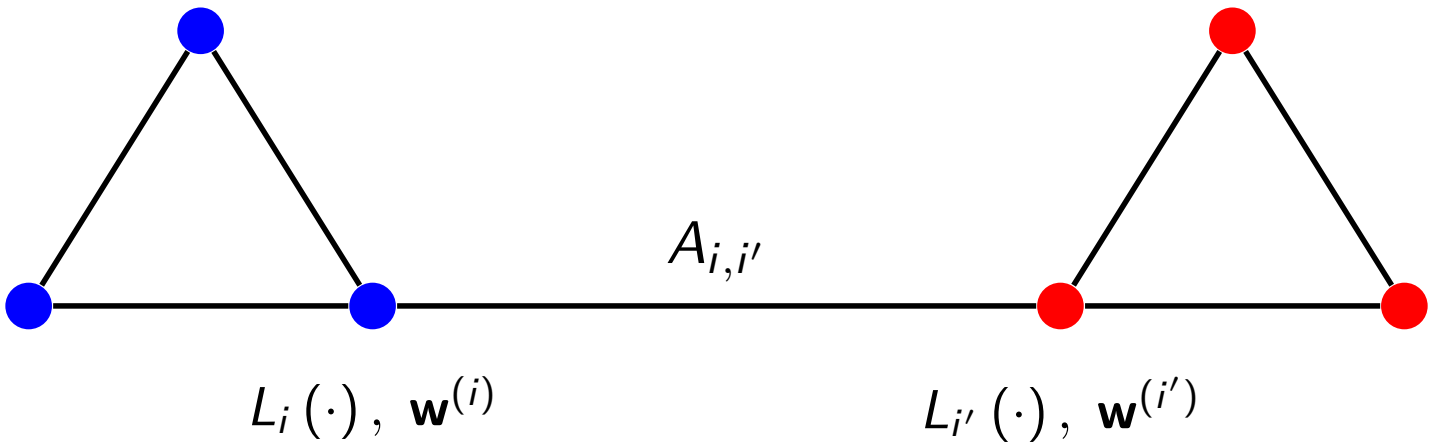
FMI Open Data

`GetFMIData.py`

`GetFMIData.py` queries hourly observations (temperature, humidity, wind speed, ...) per station.

FMI provides the *raw data* only. Everything else is **your design choice**.

From FMI Stations to an FL Network



Fixed: each device $i = 1, \ldots, n$ is a FMI station

Your design choices:

- **Data points:** define features $\mathbf{x}$ and label y from raw FMI data
- **Edges:** how to connect stations and set weights $A_{i,i'}$
- **Model:** $\mathbf{w}^{(i)} \in \mathbb{R}^d$ at each device
- **Loss:** $L_i(\cdot)$ on $\mathcal{D}^{(i)}$ (e.g. MSE)

Table of Contents

Mid-Course Feedback

Getting Started with FMI Data

The 3-Step Workflow

Validating Your Results

Deliverables and Grading

10 ECTS Extension

Next Steps

FL Design Workflow

1. **Formulate** FL application as GTVMin

2. **Characterize** solutions via fixed-point equation:

$$\mathbf{w}^{(i)} \;=\; \mathcal{F}^{(i)}\big(\mathbf{w}^{(1)}, \ldots, \mathbf{w}^{(n)}\big) \qquad \text{for each node } i$$

   where $\mathcal{F}^{(i)}$ is the **block-wise minimisation operator** at node i

3. **Iterate** the operator to obtain an FL algorithm:

$$\mathbf{w}^{(i,t+1)} \;=\; \mathcal{F}^{(i)}\big(\mathbf{w}^{(1,t)}, \ldots, \mathbf{w}^{(n,t)}\big) \qquad \text{for } t = 0, 1, 2, \ldots$$

Step 1: Formulate GTVMin

Given the FL network, solve:

$$\min_{\mathbf{w}^{(1)},\ldots,\mathbf{w}^{(n)}} \underbrace{\sum_{i=1}^{n} L_i\left(\mathbf{w}^{(i)}\right)}_{\text{local fit}} + \alpha \underbrace{\sum_{\{i,i'\}\in\mathcal{E}} A_{i,i'} \left\|\mathbf{w}^{(i)} - \mathbf{w}^{(i')}\right\|^2}_{\text{GTV penalty}}$$

- $\alpha = 0$: nodes learn **independently**
- $\alpha \to \infty$: **single shared model**
- intermediate α: **collaboration** $\propto$ **similarity**

Step 2: Fixed-Point Equation

Optimality condition for squared-ℓ_2 variation:

$$\mathbf{w}^{(i)} \;=\; \mathcal{F}^{(i)}\big(\mathbf{w}^{(1)}, \ldots, \mathbf{w}^{(n)}\big)$$

with

$$\mathcal{F}^{(i)} \;:=\; \mathbf{w}^{(i)} - \eta\Big[\nabla L_i\left(\mathbf{w}^{(i)}\right) + \alpha \sum_{i' \in \mathcal{N}^{(i)}} A_{i,i'}\left(\mathbf{w}^{(i)} - \mathbf{w}^{(i')}\right)\Big]$$

Balances local fit against agreement with neighbours.

Step 3: Iterate the Fixed-Point Operator

Initialise $\mathbf{w}^{(i,0)}$; for $t = 0, 1, 2, \ldots$ each node computes:

$$\mathbf{w}^{(i,t+1)} = \mathbf{w}^{(i,t)} - \eta \left[\nabla L_i \left(\mathbf{w}^{(i,t)} \right) + \alpha \sum_{i' \in \mathcal{N}^{(i)}} A_{i,i'} \left(\mathbf{w}^{(i,t)} - \mathbf{w}^{(i',t)} \right) \right]$$

```
for t in range(n_iter):
    for i in range(n_nodes):
        grad_local = compute_gradient(X[i], y[i], w[i])
        neighbour_diff = sum(A[i,j] * (w[i] - w[j]) for j in neighbours[i])
        w[i] = w[i] - lr * (grad_local + lam * neighbour_diff)
```

Node i only exchanges $\mathbf{w}^{(i)}$ with neighbours – no raw data shared.

Table of Contents

Mid-Course Feedback

Getting Started with FMI Data

The 3-Step Workflow

Validating Your Results

Deliverables and Grading

10 ECTS Extension

Next Steps

Sanity Checks

1. **Baseline.** Local-only training ($\alpha = 0$) – your FL method should match or improve on this.
2. **Loss curves.** Training loss should decrease monotonically (convex losses).
3. **Predictions vs. actuals.** Scatter plot for a few stations; points near the diagonal.
4. **Effect of α.** Small $\alpha \approx$ local-only; large $\alpha \approx$ shared model.
5. **Compare ≥ 2 systems.** Differ in a meaningful design choice (graph, model, α, ...).

Table of Contents

Mid-Course Feedback

Getting Started with FMI Data

The 3-Step Workflow

Validating Your Results

Deliverables and Grading

10 ECTS Extension

Next Steps

Schedule and Workflow

Milestone	Date	What you do
Kickoff	Wed 01.04	
1st submission	Sun 26.04	report (PDF) + code (zip)
Peer Review 1	27.04 – 03.05	review assigned drafts
2nd submission	Sun 17.05	revised report, response letter, talk
Peer Review 2	18.05 – 27.05	review improved reports

Component	Points
First submission	5
Peer Review 1	4
Second submission (incl. presentation)	43
Peer Review 2	8
Total	**60**

What Earns Points

- **Problem formulation.** nodes, edges, data points, loss function defined and justified
- **Methodology.** correct GTVMin and algorithm
- **Experiments.** at least 2 FL systems compared
- **Writing.** concise, follows template
- **Reproducibility.** code runs, reproduces results

Report Template and Formatting

Provided in `FLProject26/`:

- `FLReportTemplate_26.tex` – starting point (remove hints before submission)
- `FLReportSample.tex` – sample report
- `FLProjectReviewSheet.pdf` – peer review rubric

Formatting rules:

- Max 5 pages (5th page = references only)
- Anonymous – no names, GitHub links, or identifying information
- Use `ml_macros.tex` for consistent notation
- Do not modify section titles or margins

Using LLMs in Peer Review

LLMs (ChatGPT, Claude, ...) may assist your review. Two conditions:

1. **Read the report first.** Form your own assessment before using an LLM. You may be asked to explain your review.
2. **Do not upload private information to LLM.**

Good uses: check equations, look up methods, improve your review text.

Not acceptable: paste entire report $\rightarrow$ submit raw LLM output.

Table of Contents

Mid-Course Feedback

Getting Started with FMI Data

The 3-Step Workflow

Validating Your Results

Deliverables and Grading

10 ECTS Extension

Next Steps

10 ECTS – Research Project Extension

- For students aiming at a publishable report (grade ≥ 4)
- Extended scope: novel method, larger experiments, extra datasets
- Conference-paper-quality report + individual supervision
- Discuss scope **as early as possible**; work continues over summer

Interested? Email with your project idea and background.

Table of Contents

Mid-Course Feedback

Getting Started with FMI Data

The 3-Step Workflow

Validating Your Results

Deliverables and Grading

10 ECTS Extension

Next Steps

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1. **Today:** fetch FMI data, define your FL network
2. **This week:** formulate GTVMin (Step 1)
3. **Next two weeks:** implement FL algorithm (Steps 2–3), run experiments
4. **Sun 26.04:** first submission deadline

Resources

- **Report template + sample:** `FLProject26/` on MyCourses
- **FMI data script:** `GetFMIData.py` in the course GitHub repo
- **FLBook:** Ch. 4–7 for GTVMin theory and algorithms
- **Notation + figures:** The Aalto Dictionary of ML repo contains `ml_macros.tex` and 300+ reusable TikZ figures for FL networks, graphs, GTVMin, etc. – use them in your report!

GitHub: `Edition2026/assets`

GitHub: Aalto Dictionary of ML